

Atomic Theory Timeline

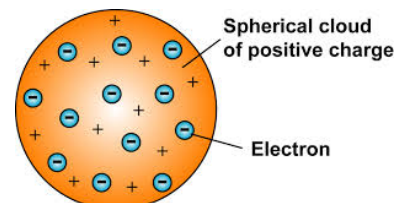
Before scientists, Greek philosophers pondered about the universe. Ancient Greeks thought that the entire universe (or the universe they knew of at the time) was made from five elements: earth, water, air, fire, and aether. Ancient Greeks thought that each of these “elements” had their natural place in the order of things: earth liked to be on the bottom, then water, then air, then fire, with aether floating on the very top. They used this idea to explain why a stone (earth) sinks in water and why fire reaches upwards in air. **Democritus**, a Greek philosopher, disagreed—and rightly so! Democritus proposed that matter could not be divided into smaller pieces forever; matter was made from small, hard particles that he called “atoms” from a Greek word meaning “indivisible”.

Fast forward over two millennia to 1785. **Charles Augustin de Coulomb** published his work on electrostatic forces, saying that opposite charges attract and the force that the charges feel is related to the amount of charge and the distance between them. His mathematical equation describing the interaction between charges is now known as “Coulomb’s law”. While this did not immediately change the idea of the atom, this did help other scientists change the atomic model.

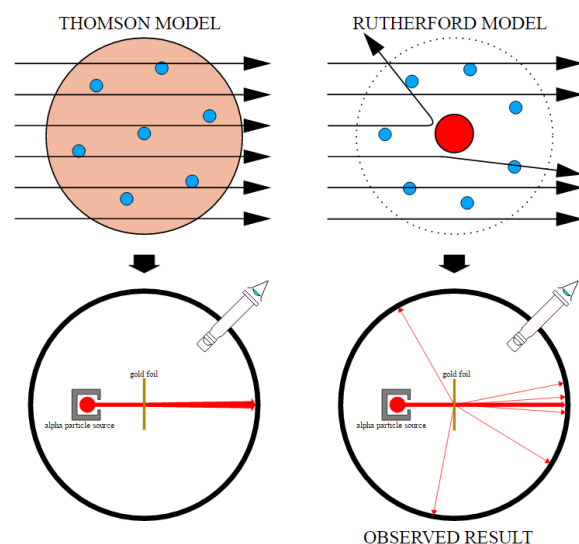
In 1808 a scientist, **John Dalton**, confirmed Democritus’ views. He stated that atoms are tiny, solid balls, like billiard balls. While he also didn’t change the idea of the atom, he introduced the first atomic theory. His theory stated:

1. Atoms are tiny, invisible particles.
2. Atoms of one element are all the same.
3. Atoms of different elements are different.
4. Compounds are formed by combining atoms.

Nearly a century later, in 1897, **J. J. Thomson** discovered electrons with cathode rays. He was the first scientist to show that the atom is made of even smaller things. So much for atoms being indivisible! This knowledge changed the atomic model to the “plum pudding model”. Thomson thought of the electrons as raisins inside a sphere of positively charged pudding.



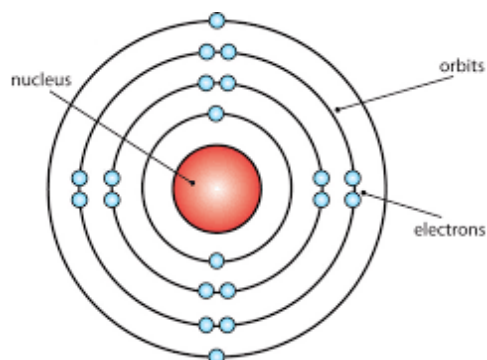
Fourteen years later, **Ernest Rutherford** published the results of his gold foil experiment (also known as “Rutherford’s gold leaf experiment”). He sent a



beam of alpha particles (which were known to be positive particles) at a thin piece of gold. Based on the plum pudding model, the alpha particles should have gone straight through the foil, at most deflecting by fractions of a degree. But what he found was very different! Most of the alpha particles did go straight through—as expected—though some were scattered by a few degrees (more than was expected) and some actually bounced backwards! Years later when commenting on this experiment, Rutherford said, “It was quite the most incredible

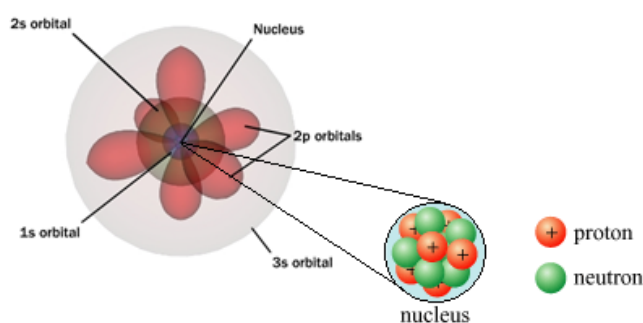
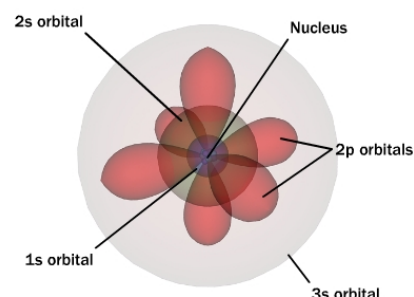
event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell [cannon ball] at a piece of tissue paper and it came back and hit you.” Rutherford couldn’t believe the results of his experiment, so he repeated it... over 100,000 times! From these experiments, Rutherford concluded that the atom must have a very dense, positive centre and most of the atom must be empty space. He called the dense centre the “nucleus” (from a Latin word meaning “little nut”) and claimed it was made from positive particles he called “protons”.

Rutherford’s model of the atom was not without flaws. The negatively charged electrons would be attracted to the positively charged nucleus (remember Coulomb’s work?). If the electrons were allowed to be anywhere on the outside of the nucleus, most of them would spiral into the centre of the atom, releasing energy as they did so. But atoms didn’t release that energy. From thinking about



the energy problems arising from Rutherford’s model, **Niels Bohr** proposed that electrons move around in specific layers or “shells”. In a 1913 publication he stated that every atom has a specific number of electron shells and thus the atomic model was modified once again to put the electrons in specific locations. This new model is sometimes referred to as the “Bohr model”, in honour of Niels Bohr, and sometimes called the “Rutherford-Bohr model”, to acknowledge both scientists’ contributions.

The next decades were a very exciting time to be involved in physics research, especially atomic theory and quantum mechanics. In 1926, **Erwin Schrödinger** published his work on the quantum wave equation, which treated electrons in atoms as waves instead of the classic billiard ball-like particles. The following year **Werner Heisenberg** published the “uncertainty principle”. One of the uncertainty principle’s forms states that you can never simultaneously know the exact location and energy of an electron. The combination of these publications lead to the idea of electron “clouds”; locations where the electrons have a high probability of being.



But there was still a problem with the model; most atomic nuclei had twice the mass they were theoretically supposed to have (based on the number of protons). Some scientists proposed that the extra mass was proton-electron pairs located in the nucleus. However, maths shows that to hold electrons in the nucleus would take way more energy than Coulomb’s law could

account for. Around the same time, other scientists discovered a type of radiation that was neutral, but it was inconsistent with gamma rays (the only then-known neutral radiation). **James Chadwick** showed that this radiation could be explained by a neutral particle. He performed numerous experiments, bombarding nitrogen, oxygen, helium, and argon with these neutral particles. From these experiments and momentum and energy laws, he was able to measure the mass of this neutral particle. In his 1932 publication, James Chadwick calls this particle the “neutron” and states that they are found in the nucleus with the protons. Since then, scientists have discovered that protons and neutrons are made up of even smaller things—quarks! But that is another story.

Simplified EAL Version

A long time ago, before modern scientists existed, Greek philosophers tried to understand the universe. They believed everything was made of five elements: **earth, water, air, fire, and aether**. They thought each element had a natural place: earth at the bottom, water above earth, air above water, fire above air, and aether at the very top. This is why they thought stones sink (made of earth) and fire rises.

One Greek philosopher, **Democritus**, disagreed. He said matter cannot be divided forever. He thought everything was made of tiny, hard particles called **atoms**, meaning “indivisible”.

Coulomb (1785)

In 1785, **Charles Augustin de Coulomb** studied electric charges. He discovered that **opposite charges attract** and the force depends on the size of the charge and the distance. His equation is called **Coulomb’s Law**. This helped later scientists understand atoms better.

Dalton (1808)

In 1808, **John Dalton** agreed with Democritus. He described atoms as **tiny, solid spheres**. He created the **first atomic theory**, which said:

- Atoms are very small and cannot be divided.
 - Atoms of the same element are identical.
 - Atoms of different elements are different.
 - Compounds form when atoms join together.
-

Thomson (1897)

In 1897, **J. J. Thomson** discovered the **electron** using cathode ray experiments. This proved atoms were not indivisible. He created the “**plum pudding model**”: electrons (negative) spread inside a ball of positive charge.

Rutherford (1911)

In 1911, **Ernest Rutherford** did the **gold foil experiment**. He shot positive alpha particles at thin gold foil. Most went through, but some bounced back. This showed:

- Atoms are mostly **empty space**.
 - Atoms have a small, dense, **positive centre** called the **nucleus**.
 - The nucleus contains **protons**.
-

Bohr (1913)

Rutherford's model could not explain why electrons did not fall into the nucleus. In 1913, **Niels Bohr** said electrons move in **fixed shells (energy levels)** around the nucleus. This is called the **Bohr model** or **Rutherford–Bohr model**.

Quantum Model (1926–1927)

In 1926, **Erwin Schrödinger** described electrons as **waves**.

In 1927, **Werner Heisenberg** said we **cannot know the exact position and energy** of an electron at the same time (the **uncertainty principle**).

These ideas led to the modern model: electrons exist in **clouds** where they are likely to be found.

Chadwick (1932)

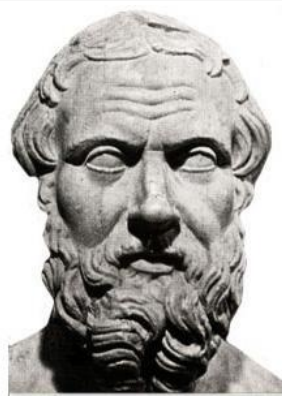
Scientists noticed the nucleus was heavier than expected. In 1932, **James Chadwick** discovered a new particle: the **neutron**, which has no charge and is found in the nucleus with protons. Later, scientists found protons and neutrons are made of even smaller particles called **quarks**.



J. J. Thomson
(1856-1940)



Niels Bohr
(1885-1962)



Democritus
(470-380 BC)



Werner Heisenberg
(1901-1976)



Erwin Schrödinger
(1887-1961)



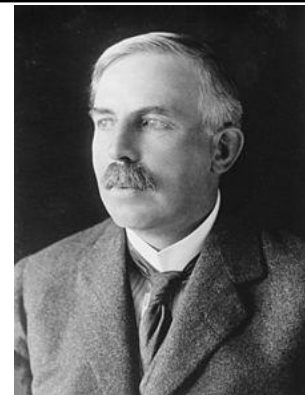
John Dalton
(1766-1844)



Charles-Augustin de Coulomb (1736-1806)



James Chadwick
(1891-1974)



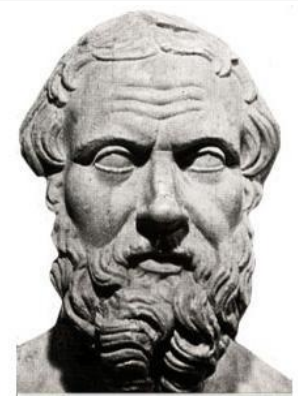
Ernest Rutherford
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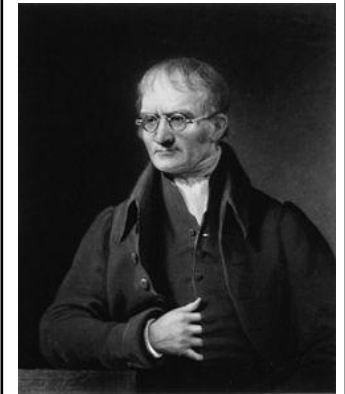
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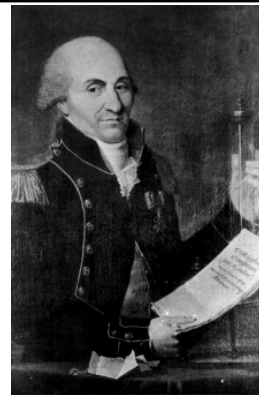
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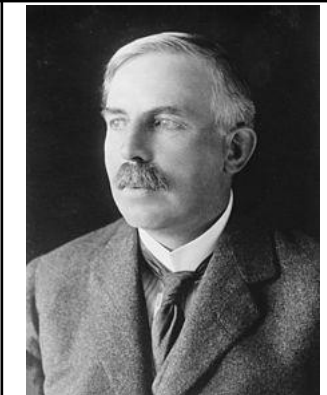
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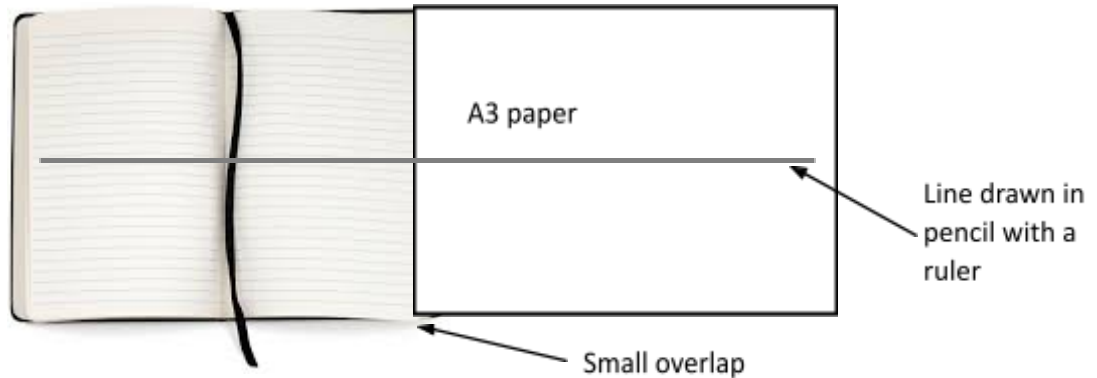
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Instructions

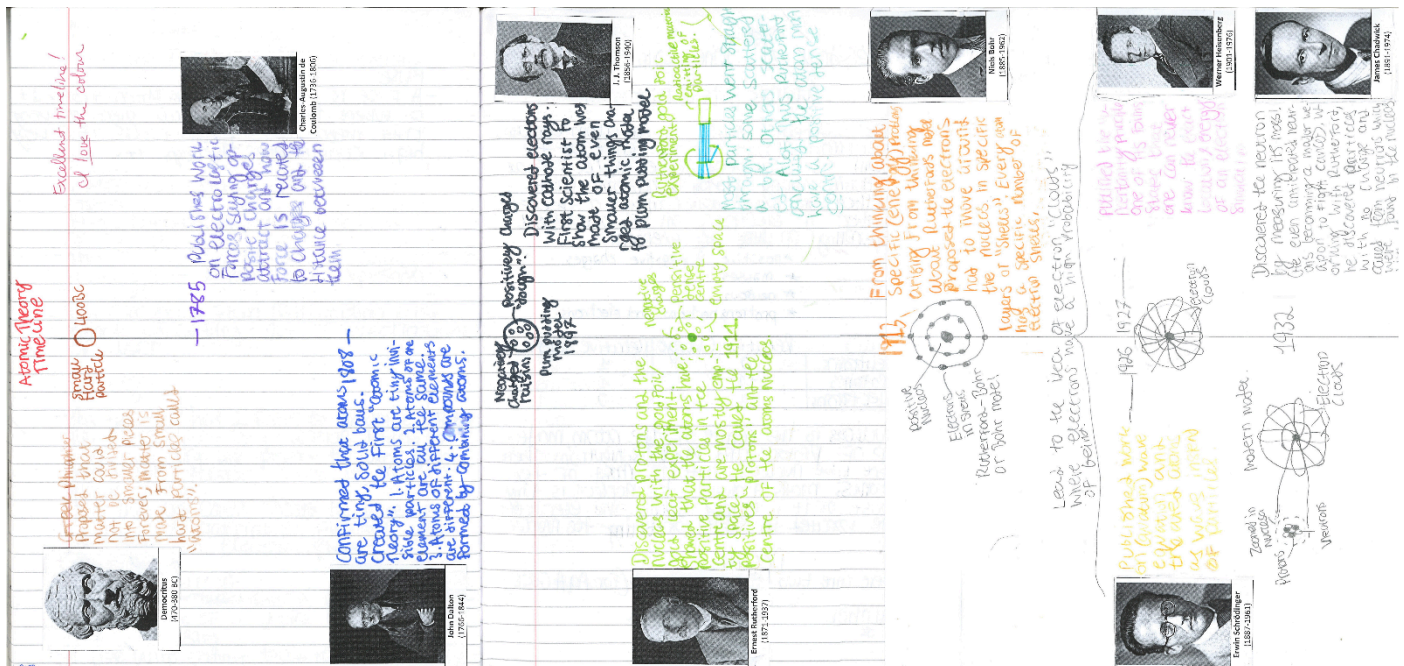
1. Open your book to a double blank page.
2. Orient the A3 paper landscape. Glue the left edge of the A3 to the right edge of your book. Overlap them just enough to ensure they stick.
3. Draw a horizontal line in pencil across the middle of your page. Afterwards, your book should look like this:



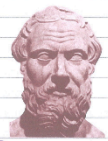
4. You may arrange your page landscape or portrait, but ensure you start on the lined paper of your book! The A3 can always be trimmed shorter.
5. Make the timeline: On the pencil line, draw the atomic model. Label the year model change (or publication leading to the model change) close to the line. The reason you drew this line in pencil is so that you can rub it out if it is going through your model. Above and below your pencil line, include information such as:
 - what the discovery was;
 - who made it (include his picture!); and
 - how the discovery was made (e.g. the experiment).

This is not an exercise in writing tiny to include every bit of detail. Make sure to include the important points and get rid of the “fluff”. Also ensure that your timeline is easy to read and appealing to study from—if you like looking at it, chances are you’ll look at it (and study it) more.

Some examples of other students' timelines:



ATOMIC THEORY TIMELINE



Democritus (470 - 380 BC)

Publishes work on electrostatic forces, saying opposite charges attract and how force is related to charges and the distance between them.



John Dalton (1766 - 1844)

Discovered electrons with cathode ray experiment. First scientist to show the atom was made of even smaller things. Proposed model to pudding model.



Ernest Rutherford (1871 - 1937)

1913 From thinking about specific (energy) problems arising from thinking about Rutherford's model, proposed the electrons had to move around the nucleus in specific layers or 'shells'. Every atom has a specific number of electron shells.



Erwin Schrödinger (1887 - 1961)



Werner Heisenberg (1901 - 1976)

Discovered the neutron by measuring its mass. Working with Rutherford he discovered particles with no charge and called them neutrons, which were found in the nucleus with the protons.

~400 BC

Small, hard particles. Atom.

Greek philosopher proposed that matter could not be divided into smaller pieces. However, matter is made from small, hard particles called 'atoms'.

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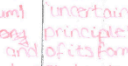
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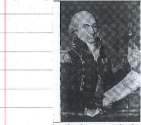


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Plum pudding model

1808
negative charged neutrons
1897
positively charged ions

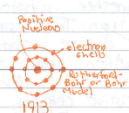


1911
Most α particles went through the foil, some were scattered a little, but some were scattered a lot! Concluded the atom must have a very dense, positive centre.



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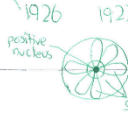
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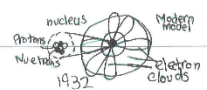
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Discovered the neutron by measuring its mass! (He even anticipated neutrons becoming a major weapon to fight cancer) Working with Rutherford, he discovered particles with no charge and called them neutrons which were found in the nucleus with the protons.

???